

Matt Ignaczak

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SUMMARY

Senior Software Engineer with 6+ years of experience building cloud-first IoT platforms on AWS. Architected the backend for a smart fire detector serving ~5K devices today, designed to scale to 290K. I own features from architecture proposal through production, and I've built the teams and mentored the engineers to ship them.

TECHNICAL SKILLS

Languages: TypeScript, Java, Kotlin, Python, SQL

Cloud (AWS): CDK, Lambda, DynamoDB, DynamoDB Streams, IoT Core, Cognito, SQS, SNS, EventBridge, CloudWatch, S3, Timestream, SES

Backend: Node.js, Serverless (Lambda), Event-Driven Architecture, Spring Boot, REST APIs, Microservices

IoT: Device Provisioning, Device Shadows, OTA Firmware Management, MQTT, BLE GATT

Mobile: Android (Jetpack Compose, MVVM, Clean Architecture), Flutter

Databases: DynamoDB, Timestream, PostgreSQL, MySQL, Firebase

Tools: Git, Azure DevOps, Jira, Postman, Mixpanel, Agentic Development

EXPERIENCE

Gentex Corporation — Senior Software Engineer

Nov 2023 – Present, Ottawa

Continued from eSight following acquisition

Place Smart Fire Detector — Cloud/IoT Platform

Core developer on the Place connected smoke/fire detector cloud platform.

- Architected the AWS cloud backend for a smart IoT fire detector (~5K devices, built to scale to 290K) using CDK, Lambda, DynamoDB, IoT Core, SQS, SNS, and EventBridge
- Built the API layer that powers the mobile app: household management, user invitations and access control, device scheduling, admin controls, and push notifications
- Designed an event-driven CDC architecture with DynamoDB Streams to decouple notifications from the API request path, improving delivery reliability
- Built the push notification pipeline end-to-end (SQS → Lambda → SNS → Firebase) for real-time fire, smoke, heat, and CO alerts with sub-second delivery via user-level MQTT topics
- Own the observability layer across 58 Lambda functions: composite alarms, concurrency alerts, DynamoDB compliance audits, and the device event pipeline feeding 100+ event types into Mixpanel
- Own the OTA firmware update infrastructure: automated job creation, S3 firmware hosting, IoT job lifecycle management across multiple device groups
- Manage the full IoT device lifecycle: provisioning, shadow state, connectivity reporting, and event schema validation
- Write architecture proposals with cost analyses and tradeoff matrices to align engineering and product on technical direction
- Led architecture deep dives on scalability (regional deployment), disaster recovery (48-hour SLA), localization, and cybersecurity
- Built production analytics dashboards for device and user metrics used by upper management for decision-making
- Coordinate across mobile, embedded, and product teams to deliver features across the full IoT stack

eSight

May 2019 – Nov 2023, Ottawa/Toronto

Software Engineer — Team Lead, Flutter Migration

2022 – 2023

- Led migration of the eSight Go companion app from native Android and iOS to Flutter, managing 2 contract developers from requirements through production
- Architected the Flutter app from scratch: project structure, navigation patterns, and state management conventions adopted as team standard
- Served as primary engineering contact for internal stakeholders, translating business requirements into technical deliverables
- Set up and maintained Azure DevOps CI/CD pipelines for build, test, and release

Software Engineer — Mobile & Platform Architecture

2020 – 2022

- Designed the software architecture for the eSight Go platform: on-device applications, BLE communication layer, and the companion mobile app
- Built the Android companion app from scratch in Kotlin with MVVM, Clean Architecture, and Jetpack Compose
- Developed Java-based Android applications running on the eSight Go HMD, owning the full on-device software stack
- Built the BLE GATT server for bidirectional communication between the glasses and companion app
- Implemented WebRTC with a BLE signalling channel for real-time data exchange between the HMD and phone
- Contributed to the eSight AWS cloud backend (Lambda)

Software Engineering Intern → Software Engineer

2019 – 2020

- Developed front-end Android features (Java) for the eSight 4 companion app across multiple production releases
- Integrated custom touchpad gesture recognition for hands-free HMD navigation
- Evaluated multiple time-of-flight sensors through comparative testing; results informed component selection for the production HMD
- Promoted to Software Engineer after 7 months

Cloud, Backend & Operations *(throughout tenure)*

- Built AWS backend infrastructure with Lambda for the HMD product line
- Developed full-stack Java application layer with Spring Boot for mobile and web clients
- Built an in-house ERP platform used by 25 production floor staff, with a reporting dashboard for upper management
- Created production analytics dashboards on AWS-hosted PostgreSQL used to identify bottlenecks and prioritize improvements
- Mentored 4 interns
- Led the company-wide GDPR compliance initiative, bringing data privacy posture up to regulatory standards across all products and internal systems
- Defined the data privacy and cybersecurity strategy; policies adopted organization-wide

EDUCATION & CERTIFICATIONS

University of Ottawa — Coursework in Computer Science, minor in Entrepreneurship & Management

AWS Certified Developer – Associate

May 2025

CIPT (Certified Information Privacy Technologist)

Sept 2024